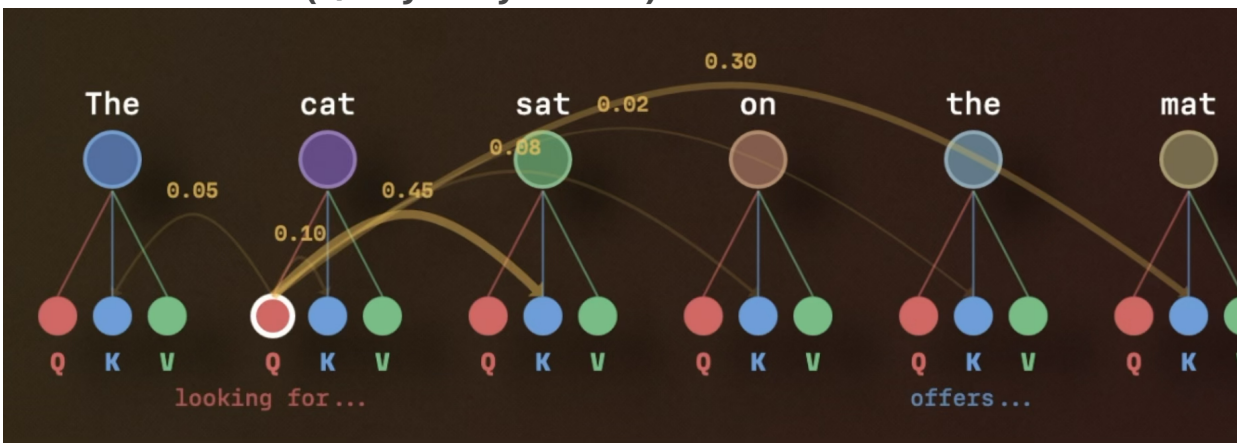
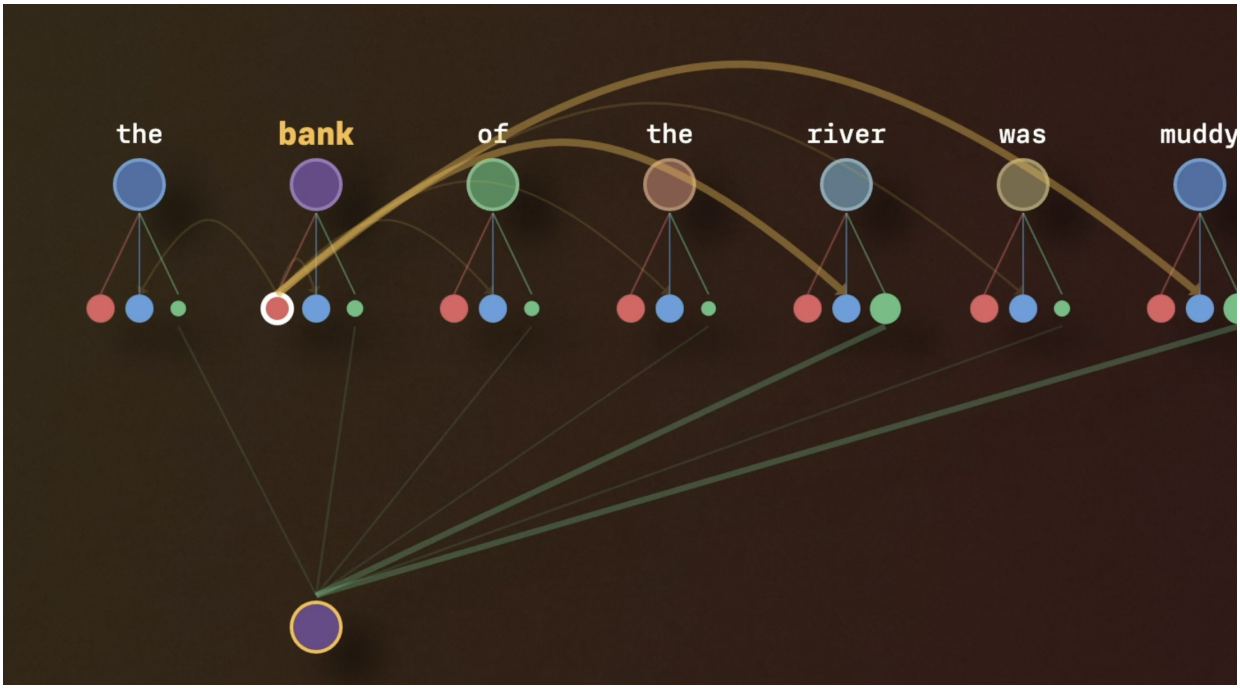
NLP Transformer Note

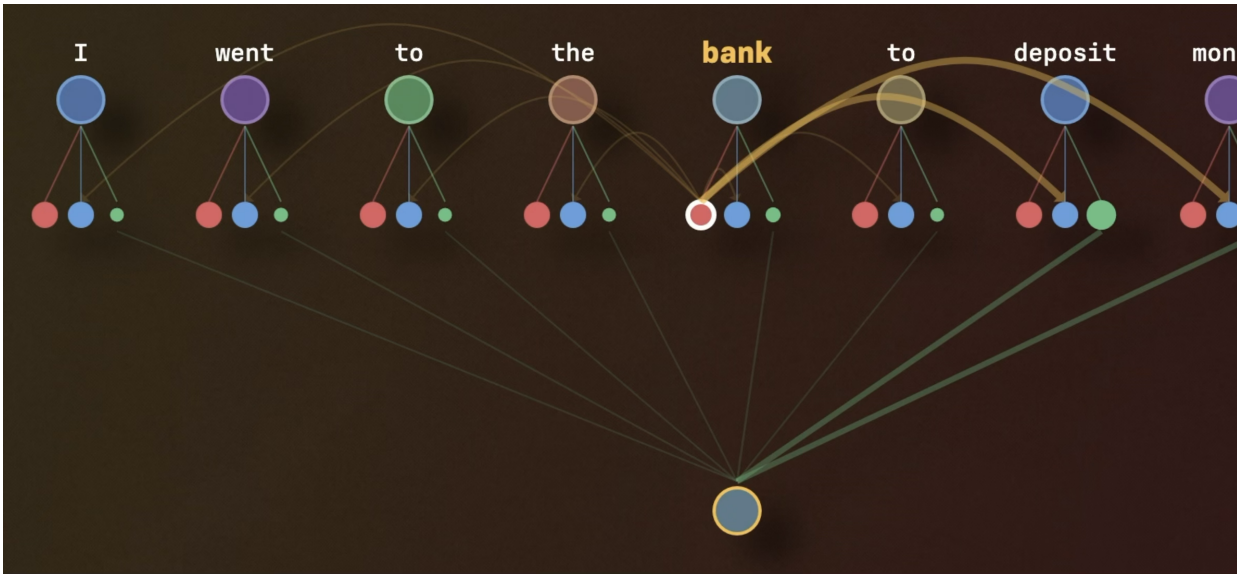
0. Self Attention (Query - Key - Value)



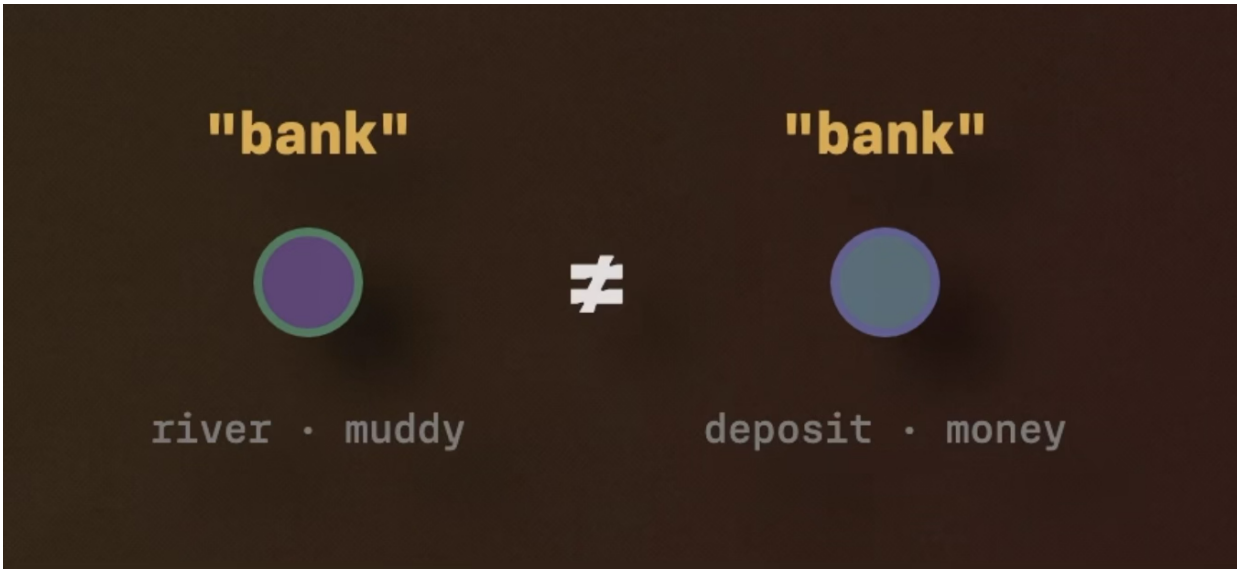
1. River context



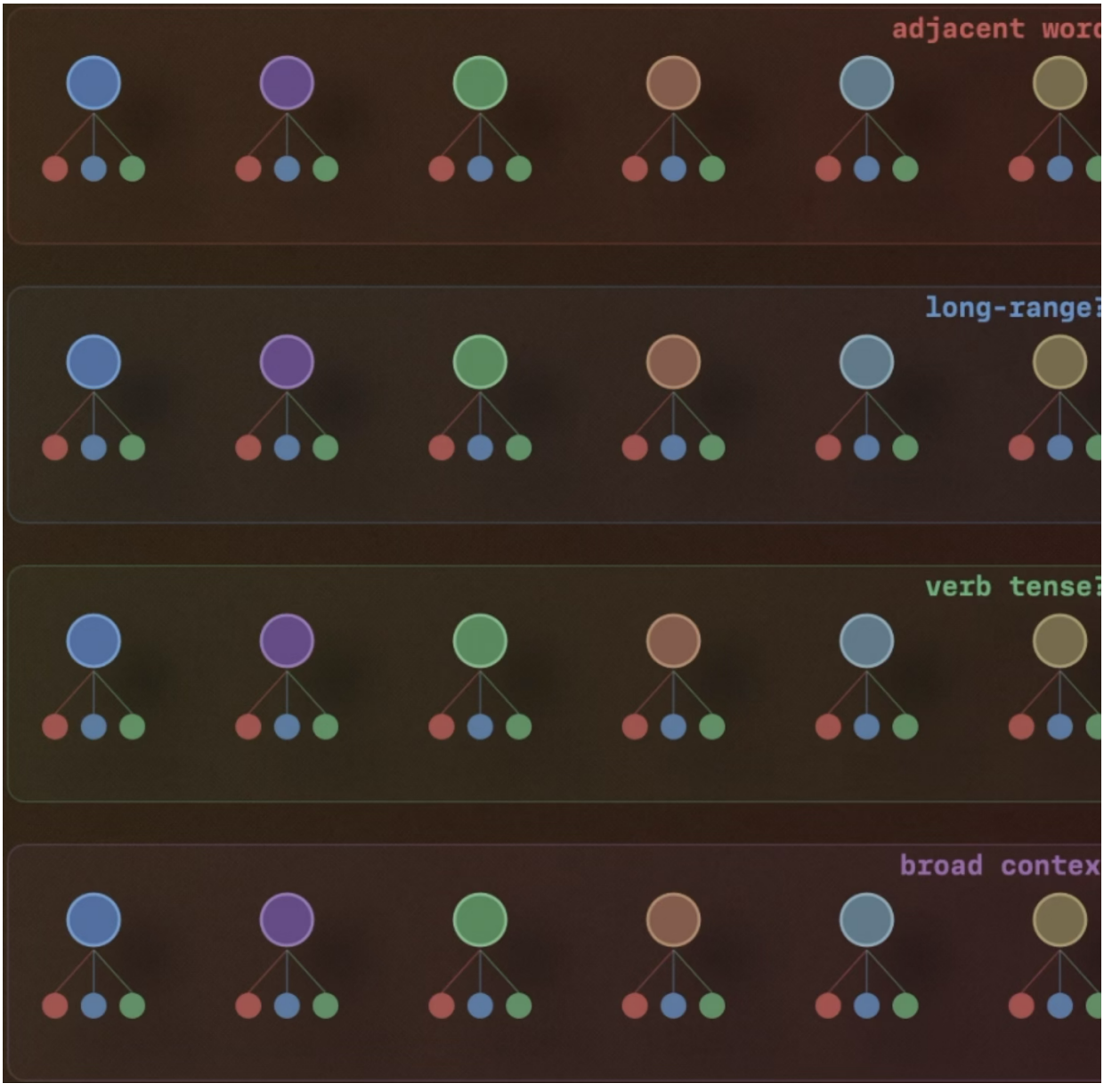
2. Finance context



Same word but meaning resolve my context. Each token's Value will mixed adjusted based on the match "Query-Key Pairs"

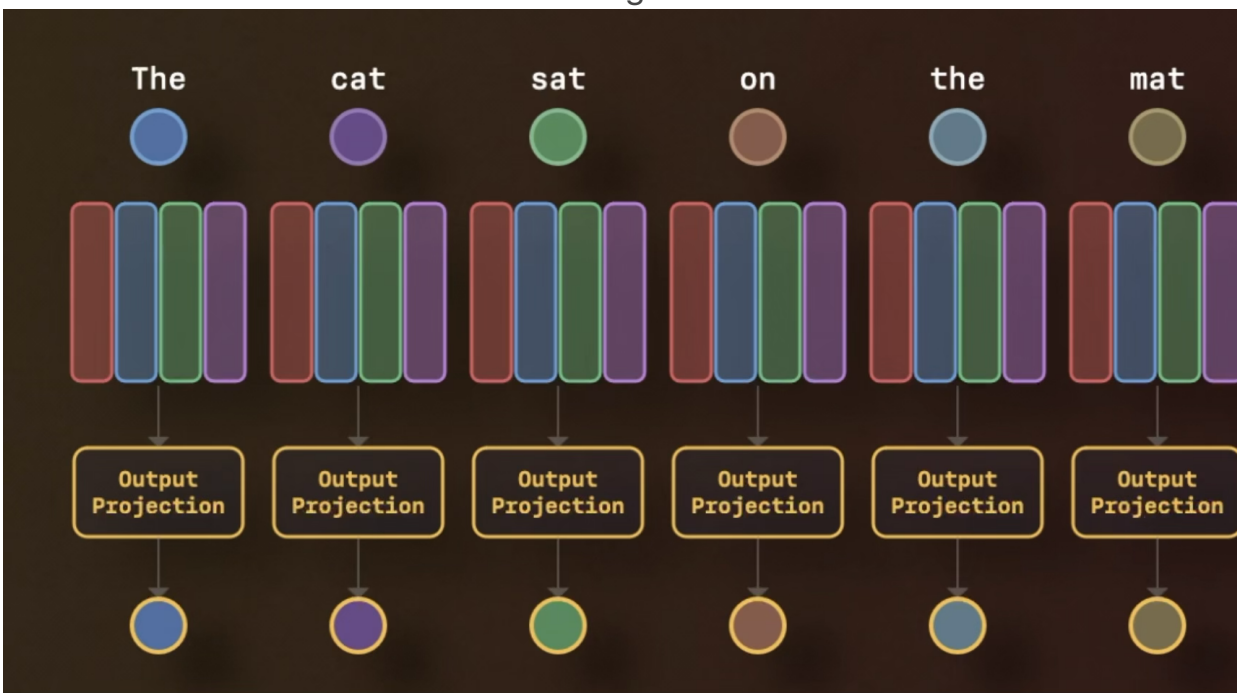


For the same token produce parallels training called "Heads" each share same token but different contexts.

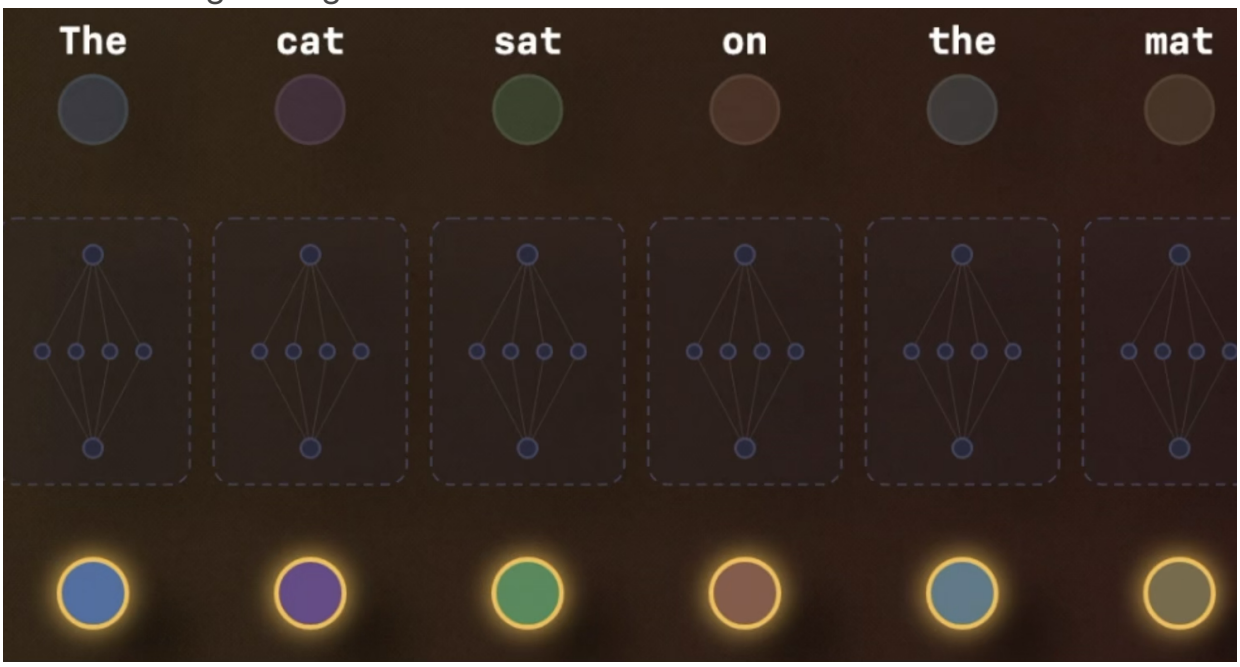


! Keys take away from pure NLP learning is “pure emergent” not “designed” => Nerual Symbolic to resolve “black box”

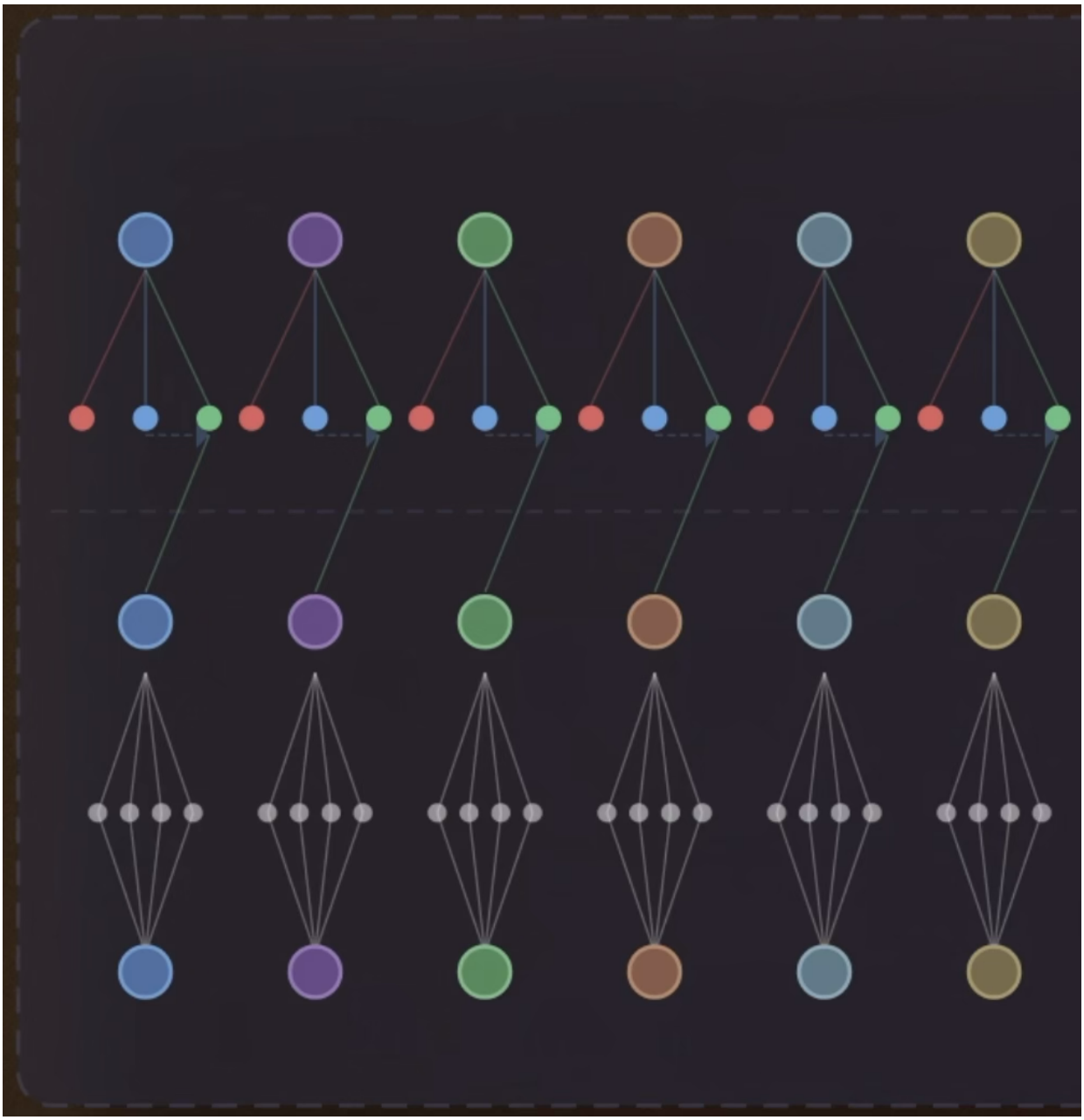
3. Theese parallels learning “Headers” will be combined back to each own token. Form “context knowledege” for each “token”
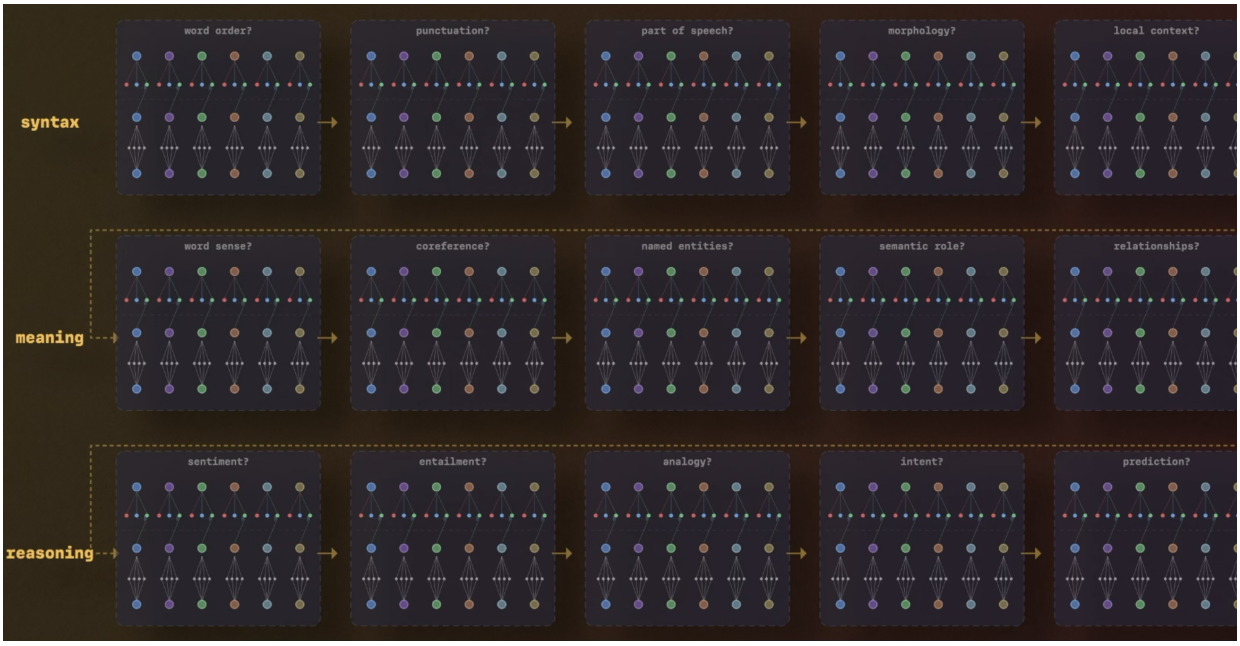


4. With these “Enrich Context” tokens will refine by simple 2-layers nerual network. Fead Forward Step helps refining these “enriched tokens” before moving next step. Avoiding noise and biases form by combining distinguish “Headers”



5. Combining “Attention” or “Heads” to “Fead Foward” form “Transformer” block or so called “Layer”



6. Stacking multiples “Transformer” blocks/layers with the input-output dimension (1 vector per token). Each “Transformer” layers build on the last.



Early layers capture basic informations. Medium layer learn the meaning of these “syntax” (basic knowledge). And Lanten Layer forming “deep reasoning”. Combine stacks of “Transformer” blocks form a “Large Language Model”.

! Noticed “Large Language Model” can abstract Deep Learning relationships of provided Data Set. But the lack the ability reasoning Why? its own output. Due to randomness of each “Self Attention” and “Feed Forwarding”. The model guessing the best weight to achieved “Target Error” But lack the Critical Reasoning to its own.